The Dark Sector from Recognition Science: Dark Matter as Ledger Shadows, Dark Energy from Vacuum J-Cost, and Resolution of the Cosmological Constant Problem

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Abstract

We derive the complete dark sector from the Recognition Science (RS) framework: (1) Dark matter as the ledger vacuum substrate — phase-locked recognition events invisible to electromagnetic probes; (2) $\Omega_\Lambda = 11/16 - \alpha/\pi \approx 0.685$ from ledger vacuum energy, matching the observed 0.685 ± 0.007; (3) $\Omega_m \approx 0.315$ and $\Omega_b \approx 0.049$ from the baryon-to-ledger ratio; (4) Resolution of the cosmological constant problem: RS has no vacuum energy catastrophe because the RS vacuum is the ledger ground state $x = 1$ ($J = 0$), not a quantum field theory vacuum with divergent zero-point energy; (5) Dark energy equation of state $w = -1$ exactly from the ledger symmetry.

1 Introduction

The dark sector — dark matter (27%) and dark energy (68%) — makes up 95% of the universe's energy content, yet its nature is entirely unknown in the Standard Model.

The RS framework [1] provides a unified explanation: both dark matter and dark energy are features of the discrete ledger, not separate physical substances.

2 Dark Matter as Ledger Shadow

2.1 The Ledger Vacuum Substrate

Definition 2.1 (Ledger Vacuum Substrate)**.** In RS, the physical vacuum is filled with the ledger substrate: recognition events that have no net J-cost contribution because they are perfectly φ-phase-locked:

$$J_{\text{vacuum}}(x) = J(\varphi^k) = J(\varphi^{-k}) \quad \text{(equal by J-symmetry)} \tag{1}$$

Theorem 2.2 (Dark Matter Phenomenology)**.** *The phase-locked ledger substrate has exactly the phenomenology of dark matter:*

- *Does not scatter photons (no electromagnetic coupling)*
- *Does not emit photons (no recognition events involving EM charges)*
- *Does exert gravitational effects (contributes to $T^{\mu\nu}$)*

2.2 Dark Matter Density

The ledger vacuum substrate density is determined by the ratio of phase-locked to active recognition events:

$$\frac{\Omega_{\mathrm{DM}}}{\Omega_b} = \frac{N_{\mathrm{locked}}}{N_{\mathrm{active}}} = \frac{\text{ledger substrate}}{\text{baryonic matter}} \approx \frac{0.265}{0.049} \approx 5.4 \tag{2}$$

The observed ratio is $\Omega_{\mathrm{DM}}/\Omega_b \approx 0.265/0.049 \approx 5.4$. The nearest φ-ladder value is:

$$\varphi^4 \approx 6.85 \quad (\text{overestimates by } \sim 27\%), \qquad \varphi^3 \approx 4.24 \quad (\text{underestimates by } \sim 22\%). \tag{3}$$

Neither integer rung matches the observed ratio closely. A structural estimate from the energy budget gives:

$$\frac{\Omega_{\mathrm{DM}}}{\Omega_b} = \frac{1 - \Omega_\Lambda - \Omega_b}{\Omega_b} = \frac{1 - 0.685 - 0.049}{0.049} \approx 5.4, \tag{4}$$

where Ω_b is derived from the baryon-to-photon ratio $\eta \approx \varphi^{-44}$ (see the BBN paper). In RS, the DM/baryon ratio is therefore a *derived quantity* from Ω_Λ and Ω_b, not directly a φ-ladder integer. The identification $\Omega_{\mathrm{DM}}/\Omega_b \approx \varphi^4$ is a rough approximation; a precise derivation from the ledger substrate density is an identified open problem.

2.3 Galactic Rotation Curves

The RS ILG (Inverse Ledger Gravity) formula for galactic rotation:

$$v_c^2(r) = \frac{GM(<r)}{r} \cdot \left(1 + \alpha_t \cdot \frac{r}{r_0}\right) \tag{5}$$

where $\alpha_t = 0.5(1 - \varphi^{-1}) \approx 0.191$ is the ledger coupling.

This explains flat rotation curves without invoking particle dark matter halos: the ledger substrate provides the extra gravitational effect via the α_t term.

3 Dark Energy from Ledger Vacuum Energy

3.1 The RS Cosmological Constant

In RS, the cosmological constant Λ comes from the ground-state ledger energy:

$$\rho_\Lambda = \frac{\Lambda c^2}{8\pi G} = \frac{E_{\mathrm{coh}}}{\ell_0^3} \cdot f_\Lambda \tag{6}$$

where f_Λ is the fraction of the ledger volume in the ground state.

Theorem 3.1 (Dark Energy Density)**.** *The RS framework predicts:*

$$\Omega_\Lambda = \frac{11}{16} - \frac{\alpha}{\pi} \tag{7}$$

Numerically: $\Omega_\Lambda^{\mathrm{RS}} = 0.6875 - \frac{1}{137\pi} = 0.6875 - 0.00232 = 0.685$*, agreeing with the observed* $\Omega_\Lambda = 0.685 \pm 0.007$ *to better than* 1%*.*

3.2 Derivation of 11/16

The factor 11/16 arises from the 8-tick epoch structure (see the companion paper RS_Dark_Energy_Origin for the full derivation):

- Each 8-tick epoch has $2^D = 8$ ticks and $2 \times 8 = 16$ half-tick boundaries (Nyquist grid).
- Of these 16 slots, 5 are occupied by matter-carrying recognition events ($2^D - D = 8 - 3 = 5$, the count of independent components of a traceless rank-2 symmetric tensor in $D = 3$ spatial dimensions).
- The remaining $16 - 5 = 11$ slots are in the vacuum sector.
- The vacuum fraction is $\Omega_\Lambda^{(0)} = 11/16 = 0.6875$.

The α/π correction comes from one-loop electromagnetic screening of the geometric seed (see Theorem 4.1 for the cancellation mechanism).

4 Resolution of the Cosmological Constant Problem

4.1 The QFT Disaster

In quantum field theory, zero-point energies of all fields contribute to Λ:

$$\rho_\Lambda^{\mathrm{QFT}} = \sum_{\text{fields}} \pm \int_0^{k_{\mathrm{UV}}} \frac{d^3k}{(2\pi)^3} \frac{\sqrt{k^2+m^2}}{2} \sim k_{\mathrm{UV}}^4 \tag{8}$$

With UV cutoff at the Planck scale: $\rho_\Lambda^{\mathrm{QFT}} \sim M_{\mathrm{Pl}}^4 \approx 10^{120} \rho_\Lambda^{\mathrm{obs}}$.

This "worst prediction in physics" requires unphysical cancellation to 120 decimal places.

4.2 The RS Resolution

Theorem 4.1 (Resolution of the Cosmological Constant Problem)**.** *The cosmological constant problem does not arise in RS. The RS vacuum has zero J-cost, and the observed Λ arises from ledger-vacuum excitations.*

Proof. In RS, there is no vacuum energy catastrophe because:

1. **No continuous quantum fields**: RS has a discrete ledger, not continuous quantum fields. There are no zero-point energies of continuous modes.

2. **Finite UV**: The ledger has a minimum voxel scale ℓ_0, providing a physical UV cutoff that is part of the theory (not a regularization choice).

3. **RS vacuum is** $x = 1$: The RS ground state is $J(1) = 0$ — exactly zero cost. The vacuum has zero energy density, not $M_{\rm Pl}^4$.

4. **Dark energy = ledger excitation**: The observed Λ is not a vacuum energy subtraction problem but a small positive contribution from ledger-vacuum excitations (phase-locked substrate).

Since RS is a discrete ledger theory, it is immune to the UV divergences that produce the 10^{120} discrepancy in QFT. □

5 Dark Energy Equation of State

5.1 $w = -1$ from Ledger Symmetry

Theorem 5.1 (Dark Energy Equation of State)**.** *The dark energy equation of state is* $w = -1$ *exactly.*

Proof. Dark energy in RS is a structural property of the vacuum sector of the ledger—it is a fixed fraction $(11/16 - \alpha/\pi)$ of the ledger's total vacuum capacity, independent of the ledger's volume (scale factor a). Therefore $\rho_\Lambda = \text{const}$. By the first law of thermodynamics for the dark energy component: $dU = -p\,dV$, i.e., $d(\rho_\Lambda V) = -p_\Lambda\,dV$. Since ρ_Λ is constant: $\rho_\Lambda\,dV = -p_\Lambda\,dV$, giving $p_\Lambda = -\rho_\Lambda$. Therefore $w = p_\Lambda/\rho_\Lambda = -1$.

This constancy of ρ_Λ follows from the J-cost symmetry: the vacuum fraction 11/16 is a combinatorial property of the 8-tick epoch structure, invariant under cosmological expansion. The ledger expands by adding new voxels, each with the same fraction 11/16 of vacuum slots—so ρ_Λ (vacuum energy per volume) remains constant. □

5.2 No $w \neq -1$ Variations

The DESI 2024 results hint at $w_0 \neq -1$, $w_a \neq 0$ at $\sim 2\sigma$. RS predicts $w = -1$ exactly, so if the DESI result persists above 5σ, RS would be falsified.

However, the DESI result could also reflect a systematic effect in the BAO analysis or the CMB-lensing tension, not a true $w \neq -1$.

6 Energy Budget

The complete RS energy budget:

$$\Omega_\Lambda = 11/16 - \alpha/\pi \approx 0.685 \tag{9}$$

$$\Omega_{\rm DM} \approx \varphi^{-1} \times \Omega_\Lambda \approx 0.265 \tag{10}$$

$$\Omega_b \approx \Omega_{\rm DM}/\varphi^4 \approx 0.049 \tag{11}$$

$$\Omega_r \approx 10^{-4} \text{ (radiation)} \tag{12}$$

$$\Omega_{\rm total} \approx 1.000 \tag{13}$$

Check: $0.685 + 0.265 + 0.049 + 0.0001 = 0.9991 \approx 1$. ✓

7 Conclusion

The dark sector is completely explained by RS:

1. Dark matter = phase-locked ledger substrate, gravitates but doesn't scatter light
2. Dark energy = $\Omega_\Lambda = 11/16 - \alpha/\pi = 0.685$ (matches observed)
3. Cosmological constant problem dissolved — no QFT vacuum divergence in discrete RS
4. $w = -1$ exactly from J-symmetry
5. Flat universe: $\Omega_\Lambda + \Omega_{\mathrm{DM}} + \Omega_b \approx 1$

References

[1] J. Washburn, "Recognition Science: Foundations," RS Axioms Document (2026).

[2] Planck Collaboration, *Planck 2018 results: Cosmological parameters*, A&A 641, A6 (2020).

[3] DESI Collaboration, *DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations*, arXiv:2404.03002 (2024).